

Shravan Ravi Narayan

Education

PostDoc, University of California, San Diego

Ph.D., University of California, San Diego

B.Eng, McGill University, Montreal

Publications

Students I advise are underlined.

1. A. Sharma, A. Balaji, Z. Yedidia, A. Du, T. Noh, I. Ireland, J. de Mooij, M. Gaudet, T. Garfinkel, D. Stefan, H. Shacham, S. Narayan. Mohabi: Disaggregating and Sandboxing the Firefox JavaScript Engine. Operating Systems Design and Implementation (OSDI), 2026.
2. T. Noh, Y. Wang, T. Garfinkel, M. Madhav, M. Erez, D. Moghimi, S. Narayan. [ARM MTE Performance in Practice](#). USENIX Security Symposium, 2026. Extended version ([Arxiv](#)).
3. S. Narayan, T. Garfinkel, E. Johnson, Z. Yedidia, Y. Wang, A. Brown, A. Vahldiek-Oberwagner, M. LeMay, W. Huang, X. Wang, M. Sun, D. Tullsen, D. Stefan. [Segue & ColorGuard: Optimizing SFI Performance and Scalability on Modern Architectures](#). Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2025.
4. S. Narayan, T. Garfinkel, M. Taram, J. Rudek, D. Moghimi, E. Johnson, C. Fallin, A. Vahldiek-Oberwagner, M. LeMay, R. Sahita, D. Tullsen, D. Stefan. [Hardware-Assisted Fault Isolation: Going Beyond the Limits of Software-Based Sandboxing](#). IEEE Micro Top Picks, 2024.
5. S. Narayan, T. Garfinkel, M. Taram, J. Rudek, D. Moghimi, E. Johnson, C. Fallin, A. Vahldiek-Oberwagner, M. LeMay, R. Sahita, D. Tullsen, D. Stefan. [Going Beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI](#). Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2023.
6. E. Johnson, E. Laufer, Z. Zhao, S. Narayan, S. Savage, D. Stefan, F. Brown. [WaVe: a verifiably secure WebAssembly sandboxing runtime](#). IEEE Symposium on Security and Privacy (S&P), 2023.
7. H. Yavarzadeh, M. Taram, S. Narayan, D. Stefan, D. Tullsen. [Half&Half: Demystifying Intel's Directional Branch Predictors for Fast, Secure Partitioned Execution](#). IEEE Symposium on Security and Privacy (S&P), 2023.
8. S. Narayan, T. Garfinkel, E. Johnson, D. Thien, J. Rudek, M. LeMay, A. Vahldiek-Oberwagner, D. Tullsen, D. Stefan. [Segue & ColorGuard: Optimizing SFI Performance and Scalability on Modern x86](#). Programming Languages and Analysis for Security (PLAS), 2022.
9. M. Kolosick, S. Narayan, C. Watt, M. LeMay, D. Garg, R. Jhala, and D. Stefan. [Isolation without taxation: Near zero cost transitions for SFI](#). Principles of Programming Languages (POPL), 2022.
10. S. Narayan, C. Disselkoen, and D. Stefan. [Tutorial: Using RLBox to sandbox unsafe C code](#). IEEE Secure Development Conference (SecDev), 2021.
11. S. Narayan, C. Disselkoen, D. Moghimi, S. Cauligi, E. Johnson, Z. Gang, A. Vahldiek-Oberwagner, R. Sahita, H. Shacham, D. Tullsen, D. Stefan. [Swivel: Hardening WebAssembly against Spectre](#). USENIX Security Symposium, 2021.
12. E. Johnson, Y. Alhessi, D. Thien, S. Narayan, F. Brown, S. Lerner, T. McMullen, S. Savage, and D. Stefan. [Доверять, но проверять: SFI safety for native-compiled Wasm](#). Network and Distributed System Security Symposium (NDSS), 2021.
13. T. Garfinkel, S. Narayan, C. Disselkoen, H. Shacham, and D. Stefan. [The road to less trusted code: Lowering the barrier to in-process sandboxing](#). Article in USENIX ;login; journal, Winter 2020.

14. S. Narayan, C. Disselkoen, T. Garfinkel, N. Froyd, E. Rahm, S. Lerner, H. Shacham, and D. Stefan. [Retrofitting fine grain isolation in the Firefox renderer](#). USENIX Security Symposium, 2020. Extended version ([Arxiv](#)).
15. J. Talpin, J. Marty, S. Narayan, D. Stefan, and R. Gupta. [Towards verified programming of embedded devices](#). Invited paper at Design, Automation and Test in Europe Conference, 2019.
16. M. Smith, C. Disselkoen, S. Narayan, F. Brown, and D. Stefan. [Browser history re:visited](#). USENIX Workshop on Offensive Technologies (WOOT), 2018.
17. F. Brown, S. Narayan, Riad S. Wahby, Dawson Engler, R. Jhala, and D. Stefan. [Finding and preventing bugs in JavaScript bindings](#). IEEE Symposium on Security and Privacy (S&P), 2017.

Others

18. Don't Get Owned by Your Dependencies: How Firefox Uses In-process Sandboxing To Protect Itself From Exploitable Libraries. Strange Loop Conference, 2022.
19. Don't Get Owned by Your Dependencies: How Firefox Uses In-process Sandboxing To Protect Itself From Exploitable Libraries. Black Hat USA Conference, 2022.
20. S. Narayan and D. Stefan. [Making software sandboxing practical using language-based techniques](#). Article in SIGPLAN PL Perspectives blog, Jul'21.
21. [RLBox: Retrofitting fine grain isolation in the Firefox renderer](#). IEEE Symposium on Security and Privacy (S&P), 2021. Invited poster.
22. S. Narayan, T. Garfinkel, S. Lerner, H. Shacham, and D. Stefan. [Gobi: WebAssembly as a practical path to library sandboxing](#). In arXiv preprint arXiv:1912.02285, Jan'19, updated Nov'19.

Awards

- 2025 Qualcomm faculty award. *Security, Memory Tagging and Sandboxing*.
- 2025 [Distinguished Reviewer](#) at ASPLOS 2025.
- 2024 Google research award. *Revisiting Memory Tagging for Security & Performance*.
- 2024 IEEE Micro Top Picks 2024. *Going beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI*.
- 2024 [Honorable mention](#), Intel Hardware Security Academic Award 2024. *Going beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI*.
- 2023 Mozilla research award. *Investigating sandboxing of JavaScript engines*.
- 2023 NSF CCF # 2327337. Collaborative Research: SaTC: CORE: Medium. *Refine the Gap: Establishing Safety for Modern Foreign Function Interfaces*. \$1,200,000 (total), \$300,000 (UT Austin).
- 2023 Most engaging speaker, UT Austin Research Topics Seminar (Course CS398T).
- 2023 [Distinguished paper award](#) at ASPLOS 2023. *Going beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI*.
- 2023 [Distinguished paper award](#) IEEE Symposium on Security and Privacy (S&P), 2023. *WaVe: a verifiably secure WebAssembly sandboxing runtime*.
- 2023 Noteworthy Reviewer Award at USENIX Security Symposium 2023.
- 2022 Mozilla research award. *RLBox v2: Extending RLBox for more performant, ubiquitous sandboxing*.
- 2022 [Winner](#), IEEE Cybersecurity Awards for Practice 2022. *Retrofitting fine grain isolation in the Firefox renderer*.
- 2022 [Honorable mention](#), NSA Best Scientific Cybersecurity Paper. *Retrofitting fine grain isolation in the Firefox renderer*.
- 2022 [Recognized contributor](#) to the Bytecode Alliance's formal verification efforts.
- 2021 [Finalist](#), Applied Research Competition, CSAW 2021. *Доверять, но проверять: SFI safety for native-compiled Wasm*.
- 2020 [Winner](#), Doctoral Award for Excellence in Research, Computer Science and Eng., UC San Diego.
- 2020 [Winner](#), Applied Research Competition, CSAW 2020. *Retrofitting fine grain isolation in the Firefox renderer*.

2020 [Distinguished paper award](#) at the USENIX Security Symposium 2020. *Retrofitting fine grain isolation in the Firefox renderer.*

Activities/service

- Dec'26–June'27 Program committee of USENIX OSDI 2027.
- Apr'26–Sep'26 Program committee for the Second cycle of ACM Computer and Communications Security 2026.
- Jan'26–Apr'26 Faculty recruiting committee at UT Austin Computer Science.
- Aug'25–Feb'26 Program committee for the Summer cycle of the ASPLOS Conference 2026.
- Nov'25–Jan'26 ACM India Doctoral Dissertation Award.
- Jan'25–Apr'25 Faculty recruiting committee at UT Austin Computer Science.
- Jan'25–Apr'25 PhD admissions committee at UT Austin Computer Science.
- Mar'25 Reviewer in the Transactions on Dependable and Secure Computing 2025.
- Mar'24–Feb'25 Program committee of the ASPLOS Conference 2025.
- Sep'24 Program committee for Posters at SOSP 2024.
- Sep'24 Program committee of the Applied Research Competition, CSAW 2024.
- Mar'24–May'24 UT Austin [Coding in the Classroom](#) outreach to introduce Programming Robots (Lego Mindstorm) to 5th grade students.
- Jan'24–Apr'24 PhD admissions committee at UT Austin Computer Science.
- Dec'23 UT Austin [Hour of code](#) outreach to introduce Programming to K12 students.
- Jun'23–Jun'24 Program committee of the USENIX Security Symposium 2024.
- Aug'23 Program committee of the Kernel Isolation, Safety, and Verification 2023.
- Jun'22–Jun'23 Program committee of the USENIX Security Symposium 2023.
- Nov'22 Program committee of the Programming Languages and Analysis for Security (PLAS), 2022.
- Nov'22 Mentor in [GradAMP](#) graduate application mentorship program at UC San Diego.
- Oct'22 Invited participant in Consumer Reports' Memory safety convening.
- Sep'22 Invited external reviewer on IEEE Symposium on Security and Privacy (S&P) 2023.
- Aug'22 Program committee of the Applied Research Competition, CSAW 2022.
- Jun'22 Program committee of the IEEE Secure Development Conference 2022.
- Sep'21–May'22 Mentor in [Early Research Scholars Program \(ERSP\)](#) undergraduate research program at UC San Diego CSE department.
- 2019–2021 Student reviewer of papers in IEEE Symposium on Security and Privacy (S&P) 2019, ACM Symposium on Operating Systems Principles 2019, USENIX Security Symposium 2020 and 2021.

Talks

- May'25 [Qualcomm Product Security Summit 2025](#) (San Diego, CA, USA). The Good, the Bad, and the Ugly: Flexibility & Performance in ARM MTE.
- Jan'25 Keynote at [Wasm Workshop 2025 \(WAW\)](#) Co-located with [POPL](#) (Denver, CO, USA). Adventures in Making Wasm Fast and More Secure.
- Oct'23 [Wasm Research Day 2023](#) (Remote). Optimizing SFI Performance, Scalability, and Spectre Resistance on Modern CPUs.
- Sept'23 WasmCon 2023 (Bellevue, WA, USA). [Don't Get Owned by Dependencies: How Firefox Uses Wasm to Protect Itself from Exploitable Libraries.](#)
- Aug'23 At RISC-V Security meeting and RISC-V Runtime Integrity SIG (Remote). Going beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI.
- June'23 [Browser Vulnerability Research Summit 2023](#) (Remote). RLBox: How Firefox Uses In-process Sandboxing To Protect Itself From Exploitable Libraries.
- May'23 KTH Royal Institute of Technology's Software Research Meetup (Remote). Retrofitting fast and secure sandboxing in real systems.

- Apr'23 At Intel Corp (Remote). Going Beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI.
- Mar'23 ASPLOS 2023 (Canada). Going Beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI.
- Mar'23 Dagstuhl Seminar: [Foundations of WebAssembly](#) (Germany). Going Beyond the Limits of SFI: Flexible and Secure Hardware-Assisted In-Process Isolation with HFI.
- Dec'22 Programming Languages and Analysis for Security (PLAS), 2022 (Remote). Segue & ColorGuard: Optimizing SFI Performance and Scalability on Modern x86.
- Nov'22 UBC's graduate security class (Remote). Swivel: Hardening WebAssembly against Spectre.
- Nov'22 CMU's security class (Remote). Retrofitting fast and secure sandboxing in real systems.
- Sept'22 Strange Loop Conference 2022 (St. Louis, MO, USA). [Don't Get Owned by Your Dependencies: How Firefox Uses In-process Sandboxing To Protect Itself From Exploitable Libraries.](#)
- Aug'22 Black Hat USA Conference 2022 (Las Vegas, NV, USA). [Don't Get Owned by Your Dependencies: How Firefox Uses In-process Sandboxing To Protect Itself From Exploitable Libraries.](#)
- June'22 At UCSD's graduate security class (San Diego, CA, USA). Retrofitting fast and secure sandboxing in real systems.
- Oct'21 IEEE Secure Development Conference 2021 (Remote). Using RLBox to sandbox unsafe C code.
- Aug'21 USENIX Security Symposium 2021 (Remote). [Swivel: Hardening WebAssembly against Spectre.](#)
- Jul'21 At Intel Corp (Remote). Swivel: Hardening WebAssembly against Spectre.
- Oct'20 MIT's CSAIL security seminar (Remote). Retrofitting fine grain isolation in the Firefox renderer.
- Oct'20 At the Center for Networked Systems' Research Review (San Diego, CA, USA). [RLBox: secure sandboxing for buggy application components.](#)
- Aug'20 USENIX Security Symposium 2020 (Remote). [Retrofitting fine grain isolation in the Firefox renderer.](#)
- Jul'20 At Brave Software, Inc (Remote). Retrofitting fine grain isolation in the Firefox renderer.

Classes

- Spring 2026 Theory and Practice of Secure Systems (CS380S)
- Fall 2025 Security and Privacy (CS361S)
- Spring 2025 Theory and Practice of Secure Systems (CS380S)
- Fall 2024 Theory and Practice of Secure Systems (CS380S)
- Fall 2023 Securing real-world systems (CS395T)

Class Guest lectures

- Spring 2025 How Spectre attacks work in "Advanced Computer Architecture (CS350C)"
- Spring 2024 Comparing WebAssembly and VMs in "Virtualization (CS360V)"
- Spring 2024 Research talk in "Intro to CS Research (CS118H)"
- Fall 2024 Research talk in "Supervised Teaching in Computer Science (CS398T)"
- Fall 2023 Research talk in "Supervised Teaching in Computer Science (CS398T)"
- Fall 2022 Research talk in "Supervised Teaching in Computer Science (CS398T)"

Software

- [rlbox](#) A general purpose library sandboxing framework that supports sandboxing plugins.
- [rlbox-wasm2c](#) An rlbox plugin that supports using libraries sandboxed with wasm2c.
- [veriwasm](#) Sandbox validator for Cranelift WebAssembly compiler.
- [swivel](#) Spectre-resistant WebAssembly compiler.